



PERSONALISED TRAINING PROGRAM

User 2

Improve cardiovascular fitness, maintain muscle mass, and support body composition through a structured 5-day training program combining strength, conditioning, and football.

36-year-old male, 73.9 kg, BMI 25.0 (Normal), 18.5% body fat (Normal), optimal muscle mass (34.5 kg SMM) and visceral fat (level 5). History of full ACL tear (left knee) and right tibia fracture — both fully resolved many years ago with excellent physical examination findings. Slightly low white cell count flagged for monitoring. Daily smoker with stress and low daily step count outside of training. Fitness level: Athletic (overall score 78, strength readiness 74, cardio readiness 76). Currently trains at the gym 4–5 days per week and plays football twice weekly. Program structure: 3 strength sessions (Push / Pull / Lower Body) + 2 football sessions replacing conditioning slots = 5 training days. Active Recovery on Tuesday, Full Rest on Friday. All exercises at Beginner + Intermediate + Advanced difficulty range per fitness scores.

PART 1 — ATHLETE INTELLIGENCE

ATHLETE PROFILE

Metric	Value	Status
Age	36	—
Gender	M	—
BMI	25.0	Normal
Body Fat % (PBF)	18.5%	Normal
Skeletal Muscle (SMM)	34.5 kg	Optimal
Visceral Fat	5.0	Optimal
Avg Steps / Day	5,556	Below Target
Avg Exercise Min / Day	0 min	Low
Resting HR	50 bpm	Athletic
Sleep Score	78	Fair
White Cell Count	3.9 x10 ³ cells/μL	Needs Attention Below

FITNESS SCORES

OVERALL FITNESS

78 / 100

Athletic

STRENGTH READINESS

74 / 100

Advanced

CARDIO READINESS

76 / 100

Advanced

Score Domain	Score	Tier	Difficulty Permitted
Strength Readiness	74	Advanced	All
Cardio Readiness	76	Advanced	All

HEART RATE ZONES

Max HR = 220 - 36 = **184 bpm** | Resting HR = **50 bpm**

Zone	% Max HR	BPM Range	Purpose
Zone 1 — Active Recovery	50-60%	92 - 110	Recovery, cooldowns
Zone 2 — Aerobic Base	60-70%	110 - 129	Fat oxidation, LDL support
Zone 3 — Aerobic Build	70-80%	129 - 147	Conditioning, circuits
Zone 4 — Threshold	80-90%	147 - 166	HIIT, sport sessions

CLINICAL CONSTRAINTS

Priority	Area	Rule
LOW	Left knee (ACL — fully resolved)	MONITOR: ACL fully resolved, excellent physical examination. PERMIT all exercises including squats, lunges, plyometrics, and football. INCLUDE single-leg stability and proprioception work as prehab in warm-up. MONITOR for any pain, swelling, or instability — flag immediately.
LOW	Right tibia fracture — fully resolved	MONITOR: Tibial fracture fully resolved, excellent physical examination. PERMIT all lower body exercises and running. MONITOR for any shin pain or discomfort during high-impact activities.
MEDIUM	White Cell Count — low (3.9×10^3 cells/ μ L)	MONITOR immune status. If User 2 presents with illness symptoms (fever, sore throat, fatigue), reduce training volume immediately. Avoid overreaching in weeks 3–4. Prioritise sleep hygiene and recovery.
MEDIUM	Daily smoking	MONITOR: Smoking impairs cardiovascular efficiency and recovery. Cap RPE at 8 during HIIT-type efforts. Ensure post-session cooldown is not skipped. Note that cardio readiness may plateau without lifestyle change.
LOW	Low daily step count (~5,556 steps)	CONSIDER: Low non-exercise activity thermogenesis (NEAT). Encourage incidental daily movement — target 8,000+ steps on non-training days. This is a low lever for body composition and cardiovascular health.

WEEKLY STRUCTURE

3 strength (Push / Pull / Lower Body) + 2 football sessions + Active Recovery + Full Rest = 7 days

Day	Type	Session	Detail
Saturday	Strength	PUSH	Chest, Shoulders, Triceps
Sunday	Sport	FOOTBALL	Football — Zone 4 cardiovascular session
Monday	Strength	PULL	Back, Biceps, Rear Delts
Tuesday	Active Recovery	ACTIVE RECOVERY	Full Body Mobility + Stretching
Wednesday	Strength	LOWER BODY	Quads, Hamstrings, Glutes, Calves
Thursday	Sport	FOOTBALL	Football — Zone 4 cardiovascular session
Friday	Full Rest	FULL REST	Complete rest day

PART 2 — TRAINING PROGRAM

SATURDAY — PUSH

Focus: Chest, Shoulders, Triceps | Location: GYM | Duration: ~55 min

WARM-UP (8 min)

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Banded Shoulder Dislocates	2	10 reps	30s	Use a wide grip and pass the band slowly over your head to open the shoulder joint. Essential prep for all pressing movements today.	Watch
Wall Slides	2	10 reps	30s	Keep your lower back, upper back, and arms in contact with the wall throughout. This activates the serratus anterior and sets the shoulder blade in a stable position before heavy pressing.	Watch
Foam Roll Thoracic Spine	1	60 seconds	15s	Cross your arms over your chest and roll your upper back only — avoid rolling the lower back. Pause on stiff segments for 5 seconds to improve thoracic extension for overhead work.	Watch
Scapular Push-Ups	2	10 reps	30s	Keep your arms completely straight — only your shoulder blades move. This activates the serratus anterior and establishes proper scapular control before the main pressing session.	Watch

MAIN BLOCK (35 min)

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Flat Barbell Bench Press	4	6–8 reps	90s	Retract your shoulder blades and maintain a slight arch in your lower back. Lower the bar to your mid-chest with control and drive explosively on the way up. Primary chest strength builder for today.	Watch
Incline Dumbbell Chest Press	3	10–12 reps	75s	Set the bench to 30–45 degrees and press the dumbbells upward in an arc, bringing them close at the top without locking out. Targets the upper chest and front delts.	Watch
Barbell Overhead Press	3	8–10 reps	90s	Brace your core hard and press the bar directly overhead — do not lean back excessively. Targets the medial and anterior deltoid with significant triceps involvement.	Watch
Cable Decline Chest Fly	3	12–15 reps	60s	Keep a slight bend in your elbows and focus on squeezing your lower chest at the peak contraction. The cable maintains tension throughout the full range — don't let it pull you back too fast.	Watch
Dumbbell Lateral Raise	3	12–15 reps	60s	Raise arms to shoulder height with a slight elbow bend — avoid swinging or using momentum. Lead with your elbows to maximise medial deltoid activation. Control the eccentric on the way down.	Watch
Cable Rope Triceps Pushdown	3	12–15 reps	60s	Keep elbows pinned to your sides throughout. Split the rope at the bottom and hold the contraction for one second before returning. Excellent triceps isolation to finish the push session.	Watch
Overhead Cable Triceps Extension	3	12–15 reps	60s	Keep your upper arms close to your head and extend fully overhead. The overhead position places the long head of the triceps under maximum stretch — valuable for complete triceps development.	Watch

CORE FINISHER (8 min)

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Hanging Knee Raise	3	12–15 reps	45s	Grip the bar with both hands and curl your pelvis upward as you raise your knees — avoid swinging. This targets the lower abdominals and hip flexors with grip strength as a secondary benefit.	Watch
Landmine Rotations	3	10 reps each side	45s	Keep arms straight and rotate from your core — not your shoulders. This is a rotational anti-rotation exercise ideal for football players who need strong obliques and trunk stability.	Watch
Forearm Plank	3	40–50 seconds	30s	Maintain a straight line from head to heels — no sagging hips. Brace as if you are about to absorb a punch to the stomach. This is fundamental core stability work.	Watch

HEART RATE STRATEGY

Max HR: **184 bpm** | Target: **Zone 2** — Aerobic Base | 60-70% = **110-129 bpm**

Anchor: Flat Barbell Bench Press

Rx: During working sets of Flat Barbell Bench Press, your HR should reach 110–129 BPM by set 3. Use the 90-second rest periods to allow HR to drop below 120 BPM before the next set. If HR exceeds 135 BPM during rest, extend rest to 2 minutes.

Note: Strength sessions target Zone 2 — this supports fat oxidation and cardiovascular base. Given User 2's low resting HR of 50 BPM (indicating good cardiac efficiency), HR elevation during rest periods should resolve quickly. If HR remains above 140 BPM throughout rest periods, reduce load by 10% on subsequent sets.

SUNDAY — FOOTBALL

Pre-Session: Perform 8–10 minutes of dynamic warm-up before the match: start with Dynamic Hip Flexor Stretch (https://youtu.be/DXuStgWuJV8?si=BPvaWKG_qsqneEHc) for 10 reps each side to open the hips, followed by Dynamic Leg Swing (<https://youtube.com/shorts/kuljYdLm7DY?feature=share>) for 10 reps each leg, then finish with Standing Toe Touch Kick (<https://youtube.com/shorts/U4oOshFWAc0?feature=share>) for 8 reps each leg to prime the hamstrings and hip flexors for sprinting and kicking demands. Hydrate well before kickoff.

Post-Session: Spend 8–10 minutes cooling down after the match: Hip Flexors to Hamstring Stretch (https://youtube.com/shorts/cKjDARUo_c8?feature=share) for 45 seconds each side to address the hip flexors and hamstrings loaded during sprinting, followed by Standing Quad Stretch (https://youtube.com/shorts/Rt45IWGdSuk?si=x9wRvV0tvA5_qDmy) for 30 seconds each side. Rehydrate with water and consume a protein source within 30–45 minutes post-match (whey protein shake recommended given supplementation history).

HEART RATE STRATEGY

Max HR: **184 bpm** | Target: **Zone 4** — Threshold | 80-90% = **147-166 bpm**

Anchor: Football

Rx: During high-intensity phases of the match (sprinting, pressing, counter-attacks), HR will naturally reach Zone 4 (147–166 BPM). During set pieces, walking phases, and half-time, allow HR to drop to Zone 2–3. Do not artificially suppress effort — the match dictates intensity.

Note: Football is classified as a Zone 4 session. Given User 2's excellent resting HR of 50 BPM and Athletic fitness level, Zone 4 efforts are well within his capacity. Given daily smoking, monitor breathing rate and fatigue during high-intensity sustained sprinting — take brief active rest if breathlessness becomes uncomfortable. Ensure White Cell Count status is considered: if User 2 is feeling unwell on match day, skip or reduce intensity.

MONDAY — PULL

Focus: Back, Biceps, Rear Delts | **Location:** GYM | **Duration:** ~55 min

WARM-UP (8 min)

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Foam Roll Thoracic Spine	1	60 seconds	15s	Cross your arms over your chest and roll through the thoracic segments. This mobilises the upper back and prepares the spine for the rowing and pulling movements in today's session.	Watch
D1 - Banded Pull Apart	2	15 reps	30s	Hold the band at shoulder height with straight arms and pull it apart, squeezing your rear delts and rhomboids at end range. This is the single best warm-up movement before any back session.	Watch
Cat-Cow	2	10 reps	20s	Move slowly and breathe — inhale as you arch (cow), exhale as you round (cat). This warms up the thoracic and lumbar spine and activates the paraspinal muscles before loaded rows.	Watch
Shoulder Mobility Circle	2	10 reps forward, 10 reps backward	20s	Make large, controlled arm circles to warm up the shoulder joint through its full range. This is essential before heavy pulling movements to ensure the rotator cuff is properly activated.	Watch

MAIN BLOCK (35 min)

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Standard Pull Up	4	6–8 reps	90s	Pull your chest to the bar and initiate the movement from your lats — not your biceps. Full dead hang at the bottom between reps. Given User 2's Athletic fitness level, bodyweight pull-ups are appropriate. Add weight via belt if 8 reps become easy.	Watch
Barbell Bent-Over Row	4	8–10 reps	90s	Hinge at hips to approximately 45 degrees, maintain a flat back, and pull the bar to your lower chest. Squeeze your shoulder blades at the top. This is the primary compound back movement of the session.	Watch
Seated Cable Row	3	10–12 reps	75s	Sit upright, pull to your navel, and hold the squeeze for one second before returning with control. Avoid rocking your torso — the movement should come entirely from your back.	Watch
Straight-Arm Cable Pulldown	3	12–15 reps	60s	Anchor the cable above head height, pull your shoulder blades back, then draw your straight arms down toward your thighs. This isolates the lats and trains the crucial shoulder extension pattern used in sprinting and change of direction.	Watch
Rear Delt Fly Machine	3	12–15 reps	60s	Keep your back straight and use a controlled motion — avoid swinging. Squeeze your shoulder blades together at the end of each rep. Rear delt strength is essential for shoulder health, posture, and sports performance.	Watch
EZ Bar Curl	3	10–12 reps	60s	Use the angled grip to reduce wrist strain. Keep elbows pinned at your sides and curl with full range — full extension at the bottom, full contraction at the top. Avoid swinging.	Watch
Bicep Hammer Curl	3	10–12 reps	60s	Keep palms facing each other throughout the entire movement. The neutral grip targets the brachialis and brachioradialis in addition to the biceps, building overall arm thickness.	Watch

CORE FINISHER (8 min)

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Ab Wheel Rollout	3	8–10 reps	45s	Keep your core braced and avoid letting your lower back sag at the end range of the rollout. This is an advanced anti-extension core exercise — if you cannot control the lower back, reduce the range of motion.	Watch
Side Plank	3	35–45 seconds each side	30s	Stack your feet and keep your hips lifted in a straight line from head to heel. This trains lateral core stability — critical for the deceleration and change-of-direction demands of football.	Watch
D1 - Bird Dog	3	10 reps each side	30s	Extend the opposite arm and leg simultaneously while keeping your hips level and lower back neutral. Hold each extension for 2 seconds before returning. Excellent for lumbar stability and motor control.	Watch

HEART RATE STRATEGY

Max HR: **184 bpm** | Target: **Zone 2** — Aerobic Base | 60-70% = **110-129 bpm**

Anchor: Standard Pull Up

Rx: During sets of Standard Pull Ups, HR should reach 110–129 BPM by the third set. Use 90-second rest periods to allow HR to recover below 120 BPM before the next set. Monitor effort on the compound barbell row — HR may spike briefly to Zone 3, which is acceptable.

Note: Pull sessions are Zone 2 strength work. Given User 2 had football yesterday, monitor for residual fatigue. If HR is elevated at rest (above 65 BPM on arrival), consider reducing the load on the first two exercises by 10% to account for accumulated fatigue from Sunday's football session.

TUESDAY — ACTIVE RECOVERY

Focus: Full Body Mobility + Stretching | **Location:** GYM | **Duration:** ~35 min

MOBILITY FLOW

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Foam Roll Thoracic Spine	1	90 seconds	none	Cross your arms over your chest and roll slowly through the upper back. Pause on any stiff or tender segments for 10–15 seconds. Prioritise thoracic mobility after three consecutive training days.	Watch
Foam Roll Quads	1	90 seconds each leg	none	Roll slowly from hip to knee and pause on tender spots for 20–30 seconds. The quads and hip flexors are heavily loaded during football — this aids recovery and reduces muscle soreness.	Watch
90/90 Hip Rotations	2	10 reps each side	20s	Sit with both legs at 90 degrees and rotate your hips side to side, lowering the forward knee toward the floor. Excellent for hip capsule mobility — particularly important for User 2 given his resolved ACL history.	Watch
Pigeon Pose	2	45 seconds each side	20s	Work into the front hip external rotation and breathe deeply into the stretch. This is one of the most effective stretches for the piriformis and hip external rotators — essential for football players.	Watch
Dynamic Hip Flexor Stretch	2	10 reps each side	20s	Try to increase the range of motion each time. The hip flexors are chronically shortened in footballers due to repeated kicking and sprinting — consistent stretching is key to preventing anterior hip impingement.	Watch
Hip Flexors to Hamstring Stretch	2	45 seconds each side	20s	Alternate between a hip flexor lunge and a hamstring stretch smoothly. This compound stretch addresses the two most commonly loaded muscle groups in football in a single fluid movement.	Watch
Thread-the-Needle	2	10 reps each side	20s	Reach one arm under your body and rotate your upper back as far as comfortable. This restores thoracic rotation — critical after heavy rows and bench pressing earlier in the week.	Watch
Quadruped Thoracic Rotation	2	10 reps each side	20s	Place one hand behind your head and rotate upward, opening your chest fully. Excellent thoracic rotation exercise that complements the thread-the-needle.	Watch
Ankle Dorsiflexion Rockers	2	15 reps each ankle	15s	Rock forward over your toes keeping your heel on the ground. Given User 2's history of right tibia fracture, maintaining good ankle dorsiflexion is important to ensure proper force distribution through the lower limb during sport.	Watch
Deep Squat Hold	2	45 seconds	20s	Sit deep into the squat with chest up and heels on the ground. Use a doorframe or rack for support if needed. This restores ankle dorsiflexion range, hip mobility, and thoracic extension simultaneously.	Watch
D1 - Single Leg Balance	2	30 seconds each leg	15s	Stand on one leg and focus on a fixed point. Given User 2's resolved left ACL, regular single-leg balance work is excellent prehab to maintain neuromuscular control and reduce re-injury risk. Progress to eyes closed over time.	Watch
Child's Pose with Side Reach	2	45 seconds each side	20s	From a lunge position, gently drive the hips forward to stretch the hip flexor, then sit the hips back and straighten the front knee to stretch the hamstring. A comprehensive lower body stretch to finish the mobility session.	Watch

HEART RATE STRATEGY

Max HR: **184 bpm** | Target: **Zone 1** — Active Recovery | 50-60% = **92-110 bpm**

Anchor: 90/90 Hip Rotations

Rx: Heart rate should remain between 92–110 BPM throughout the entire active recovery session. If any movement causes HR to exceed 110 BPM, slow down or take a seated rest. This session is restorative — zero intensity is the goal.

Note: Zone 1 is non-negotiable for active recovery. This session sits between Saturday Push, Sunday Football, and Monday Pull — it is the most important recovery window of the week. User 2's sleep score of 78 is fair; prioritise this session and ensure it is not skipped or converted into additional training.

WEDNESDAY — LOWER BODY

Focus: Quads, Hamstrings, Glutes, Calves | **Location:** GYM | **Duration:** ~55 min

Warning: MONITORING NOTE: User 2 has a fully resolved left ACL and right tibia fracture. All lower body exercises are appropriate. Monitor for any knee pain, instability, or tibial discomfort during loading. Include prehab stability work in warm-up as standard. No exercises excluded.

WARM-UP (8 min)

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Dynamic Hip Flexor Stretch	2	10 reps each side	20s	Drive the hips forward in the lunge position and increase range each rep. Essential warm-up for any lower body session, particularly before squatting and hip hinging.	Watch
Lateral Band Walk	2	15 steps each direction	30s	Keep tension in the band and stay in a half-squat position. This activates the glutes and hip abductors, setting up the hips for stable squatting and Romanian deadlift patterns today.	Watch
Goblet Squat	2	10 reps (light)	30s	Hold a light dumbbell close to your chest and sit deep between your legs. This is a movement prep drill — use a load light enough to move freely through full range. Prime the quads, glutes, and ankle dorsiflexion.	Watch
Dynamic Leg Swing	2	10 reps each leg	20s	Swing your leg forward and backward in a controlled pendulum motion. This warms up the hip flexors, hamstrings, and hip joint specifically — critical before the deadlift pattern in today's session.	Watch

MAIN BLOCK (35 min)

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Barbell Back Squat	4	6–8 reps	2 min	Keep your chest up, push your knees out in line with your toes, and sit back into your hips. Given User 2's resolved ACL, control the descent and avoid excessive forward knee travel. This is the primary lower body strength exercise of the week.	Watch
Romanian Deadlift (Barbell)	4	8–10 reps	90s	Push your hips back and keep the bar close to your legs throughout. Feel a deep stretch in the hamstrings at the bottom and drive through your hips to lockout. Essential posterior chain exercise for football performance.	Watch
Hack Squat Machine	3	10–12 reps	75s	Place your feet shoulder-width apart and press through your whole foot. The hack squat places the quad under significant load with reduced spinal loading — excellent supplemental quad work after the barbell squat.	Watch
Hip Thrust	3	12–15 reps	75s	Drive through your heels and squeeze your glutes hard at the top — hold for one second. The hip thrust is the most effective glute-building exercise and directly transfers to sprinting acceleration in football.	Watch
Hamstring Leg Curl	3	12–15 reps	60s	Control the lowering phase — take 2–3 seconds on the eccentric. Avoid letting the weight drop. Hamstring strength is critical for knee stability and injury prevention, particularly for former ACL patients.	Watch
Leg Press Single-Leg	3	10–12 reps each leg	60s	Press through your heel and keep your knee tracking over your toes. Single-leg loading ensures symmetry between the left and right leg — important for User 2 to maintain balanced strength around his previously injured left knee.	Watch
Donkey Calf Raise	3	15–20 reps	45s	Get a full stretch at the bottom and rise fully onto your toes at the top. The donkey calf raise places the gastrocnemius under maximum load due to the bent-over position. Calf strength aids ankle stability and sprint performance.	Watch

CORE FINISHER (8 min)

Exercise	Sets	Reps / Time	Rest	Coaching Notes	Video
Hanging Leg Raise	3	10–12 reps	45s	Keep your legs straight and raise until parallel to the floor — avoid swinging. This is an advanced core exercise appropriate for User 2's Athletic fitness level. Engages the entire anterior chain.	Watch
Russian Twist With Ball	3	20 reps (10 each side)	45s	Rotate from your torso — not just your arms — and keep your feet off the ground for maximum difficulty. Rotational core strength directly benefits kicking power and change-of-direction in football.	Watch
D1 - Pallof Press	3	10 reps each side	30s	Press the cable straight out and resist rotation with your core — the further out you press, the harder the anti-rotation challenge. This directly trains the core stability needed to absorb contacts and change direction in football.	Watch

HEART RATE STRATEGY

Max HR: **184 bpm** | Target: **Zone 2** — Aerobic Base | 60-70% = **110-129 bpm**

Anchor: Barbell Back Squat

Rx: During heavy barbell back squat sets, HR should reach 110–129 BPM by the third set. Use the full 2-minute rest period between squat sets. On accessory exercises (leg curl, calf raise), HR may drop to Zone 1–2 — this is normal for lower-intensity isolation work.

Note: Lower body sessions with heavy compound movements may briefly push HR to Zone 3 — this is acceptable. Monitor recovery between sets. Given User 2 smokes daily, if he notices delayed breathing recovery during heavy squat sets, extend rest to 2 min 30 sec and reduce load slightly until cardiovascular adaptation improves.

THURSDAY — FOOTBALL

Pre-Session: Perform 8–10 minutes of dynamic warm-up: begin with Standing Hip Circle (https://youtube.com/shorts/m_J-Hia1XVk?feature=share) for 10 circles each direction per leg to loosen the hip joint, then Dynamic Hamstring Sweep Stretch (<https://youtube.com/shorts/pc2Z9Z9CpFg?feature=share>) for 10 reps to prime the hamstrings before sprinting, finishing with Butt Kicks (<https://youtube.com/shorts/Bugm-uCTILE?feature=share>) for 30 seconds to raise core temperature and activate the knee flexors. User 2 has had a full lower body strength session on Wednesday — this warm-up is especially important to ensure full muscle readiness.

Post-Session: Perform 8–10 minutes of cool-down: Standing Calf Stretch (https://youtube.com/shorts/nx9RVV9TC_w?si=Kubk1exeAhHPhEN_) for 30 seconds each side to address the significant calf demand from running and jumping, followed by Supine Twist (<https://youtube.com/shorts/EIKoMMaTPCM?si=D0bdNkz9T83Mhvtf>) for 30 seconds each side to release spinal compression from running. Rehydrate promptly and consume whey protein within 45 minutes. Tomorrow is Full Rest — use it fully.

HEART RATE STRATEGY

Max HR: **184 bpm** | Target: **Zone 4** — Threshold | 80-90% = **147-166 bpm**

Anchor: Football

Rx: High-intensity phases of football (pressing, sprinting, counter-attacks) will naturally reach Zone 4 (147–166 BPM). During set pieces and walking phases, allow HR to drop to Zone 2–3. Note: playing football the day after a heavy lower body session may produce initial leg fatigue — this is normal and should resolve within the first 10 minutes of warm-up.

Note: This is User 2's second football session of the week. Given the lower body session was Wednesday, monitor for quadriceps or hamstring fatigue that may affect sprint quality. If significant soreness is present, communicate with the coach to manage User 2's minutes or positional demands. White Cell Count flag: if User 2 presents with fatigue or illness symptoms, postpone the session.

FRIDAY — FULL REST

Complete rest. No structured activity. Prioritise sleep, hydration, and nutrition. This rest day follows two consecutive high-intensity football and training days — it is a critical recovery window. User 2 should aim for 7–9 hours of sleep. Avoid alcohol and prioritise whole food nutrition.

Focus on protein intake (1.6–2.0g per kg of bodyweight — approximately 118–148g protein daily at 73.9 kg) to support muscle protein synthesis from the week's training. Given User 2 takes whey protein, add one serving post-training on training days and consider a casein or whole-food protein source before sleep to maximise overnight recovery. Multi-vitamin supplementation is on track — ensure it is taken with a meal for optimal absorption. Daily smoking increases oxidative stress and vitamin C demand — consider whole food sources high in antioxidants (citrus, berries, leafy greens) on rest days.

WEEKLY OVERVIEW — ALL EXERCISES

Saturday — PUSH (55 min)

Warm-Up: Banded Shoulder Dislocates (2×10 reps) • Wall Slides (2×10 reps) • Foam Roll Thoracic Spine (1×60 seconds) • Scapular Push-Ups (2×10 reps)

Main: Flat Barbell Bench Press (4×6–8 reps) • Incline Dumbbell Chest Press (3×10–12 reps) • Barbell Overhead Press (3×8–10 reps) • Cable Decline Chest Fly (3×12–15 reps) • Dumbbell Lateral Raise (3×12–15 reps) • Cable Rope Triceps Pushdown (3×12–15 reps) • Overhead Cable Triceps Extension (3×12–15 reps)

Core: Hanging Knee Raise (3×12–15 reps) • Landmine Rotations (3×10 reps each side) • Forearm Plank (3×40–50 seconds)

Sunday — FOOTBALL

Football — see pre/post session notes on dedicated page

Monday — PULL (55 min)

Warm-Up: Foam Roll Thoracic Spine (1×60 seconds) • D1 - Banded Pull Apart (2×15 reps) • Cat-Cow (2×10 reps) • Shoulder Mobility Circle (2×10 reps forward, 10 reps backward)

Main: Standard Pull Up (4×6–8 reps) • Barbell Bent-Over Row (4×8–10 reps) • Seated Cable Row (3×10–12 reps) • Straight-Arm Cable Pulldown (3×12–15 reps) • Rear Delt Fly Machine (3×12–15 reps) • EZ Bar Curl (3×10–12 reps) • Bicep Hammer Curl (3×10–12 reps)

Core: Ab Wheel Rollout (3×8–10 reps) • Side Plank (3×35–45 seconds each side) • D1 - Bird Dog (3×10 reps each side)

Tuesday — ACTIVE RECOVERY (35 min)

Mobility: Foam Roll Thoracic Spine (1×90 seconds) • Foam Roll Quads (1×90 seconds each leg) • 90/90 Hip Rotations (2×10 reps each side) • Pigeon Pose (2×45 seconds each side) • Dynamic Hip Flexor Stretch (2×10 reps each side) • Hip Flexors to Hamstring Stretch (2×45 seconds each side) • Thread-the-Needle (2×10 reps each side) • Quadruped Thoracic Rotation (2×10 reps each side) • Ankle Dorsiflexion Rockers (2×15 reps each ankle) • Deep Squat Hold (2×45 seconds) • D1 - Single Leg Balance (2×30 seconds each leg) • Child's Pose with Side Reach (2×45 seconds each side)

Wednesday — LOWER BODY (55 min)

Warm-Up: Dynamic Hip Flexor Stretch (2×10 reps each side) • Lateral Band Walk (2×15 steps each direction) • Goblet Squat (2×10 reps (light)) • Dynamic Leg Swing (2×10 reps each leg)

Main: Barbell Back Squat (4×6–8 reps) • Romanian Deadlift (Barbell) (4×8–10 reps) • Hack Squat Machine (3×10–12 reps) • Hip Thrust (3×12–15 reps) • Hamstring Leg Curl (3×12–15 reps) • Leg Press Single-Leg (3×10–12 reps each leg) • Donkey Calf Raise (3×15–20 reps)

Core: Hanging Leg Raise (3×10–12 reps) • Russian Twist With Ball (3×20 reps (10 each side)) • D1 - Pallof Press (3×10 reps each side)

Thursday — FOOTBALL

Football — see pre/post session notes on dedicated page

Friday — FULL REST

Complete rest. No structured activity. Prioritise sleep, hydration, and nutrition. This rest day follows two consecutive high-intensity football and training days — it is a critical recovery window. User 2 should aim for 7–9 hours of sleep. Avoid alcohol and prioritise whole food nutrition.

PERFORMANCE NOTES

Low White Cell Count (3.9×10^3 cells/ μ L)

Implication: White cell count is mildly below the normal range, which may indicate reduced immune surveillance. High training loads, lack of sleep, and chronic smoking all suppress immune function independently — the combination warrants monitoring. User 2 may be more susceptible to infections, and overreaching could further suppress immunity.

Action: Monitor for signs of illness (fatigue beyond normal DOMS, sore throat, elevated resting HR > 5–10 BPM above normal). If resting HR is elevated on any given morning, reduce that session's intensity by 20%. Do not train if acutely unwell. Schedule a follow-up blood test in 6–8 weeks to confirm whether WBC has normalised. Consult GP if WBC remains low.

Daily Smoking

Implication: Smoking significantly impairs cardiovascular efficiency, lung capacity, and recovery rate. Carbon monoxide from cigarette smoke reduces oxygen-carrying capacity of the blood, meaning User 2's effective aerobic capacity is lower than his fitness scores suggest. Post-exercise recovery is slower, and the risk of respiratory illness is elevated.

Action: Ensure full warm-up and cool-down are never skipped — the respiratory system needs more time to prepare and recover. During conditioning blocks within football sessions, if breathlessness becomes uncomfortable (unable to speak in short sentences), reduce intensity and adopt Zone 3 rather than Zone 4 efforts. Cessation or significant reduction of smoking will produce measurable fitness improvements within 4–8 weeks — strongly recommend discussing smoking cessation support with a GP.

Low Daily Step Count (~5,556 steps)

Implication: User 2's non-exercise physical activity (NEAT) is low, suggesting significant sedentary periods outside of structured training and football. NEAT accounts for a meaningful portion of daily caloric expenditure and influences cardiovascular health, insulin sensitivity, and recovery. Low NEAT may blunt some of the body composition benefits of training.

Action: Target a minimum of 8,000 steps per day — on training days and rest days. Simple strategies: walk to locations within 20 minutes on foot instead of driving, take the stairs, stand during phone calls. A cheap intervention with meaningful impact on health, particularly given the low HDL-adjacent risks of a sedentary lifestyle and smoking combination.

Resolved ACL (Left Knee) + Right Tibia Fracture — Prehab Integration

Implication: Both injuries are fully resolved with an excellent physical examination. However, long-term ACL repair carries a small but elevated risk of re-tear — particularly with the high-intensity change of direction, jumping, and contact demands of football. Similarly, the right tibia fracture site should be monitored under high-impact loading.

Action: ACL prehab exercises are embedded in the weekly warm-ups and active recovery: single-leg balance, lateral band walks, goblet squat. These should be performed consistently — do not skip warm-up components. Report any new knee swelling, locking, giving way, or tibial pain immediately to the physio. Consider scheduling a physio check-up every 3–6 months given the high football volume.

PROGRESSION GUIDE

Weeks 1-2 — Foundation

Foundation phase. Use the prescribed rep ranges at a load that leaves 2–3 reps in reserve (RPE 7–8). Focus on movement quality, warm-up execution, and consistency across all 5 training days. Do not attempt personal bests in weeks 1–2 — this is a baseline-setting period. Cap RPE at 8 on all compound lifts. Ensure active recovery on Tuesday is not skipped or rushed. Monitor resting HR daily if possible — if it is elevated by more than 5–7 BPM above baseline, reduce volume that day. Note any left knee or right shin discomfort during lower body sessions and flag immediately.

Weeks 3-4 — Load

Load progression phase. Increase load on major compound lifts (Barbell Back Squat, Romanian Deadlift, Flat Barbell Bench Press, Standard Pull Up) by 2.5–5 kg where RPE was below 8 in weeks 1–2. Alternatively, add one additional set to one compound exercise per session per week. On conditioning-type football sessions, monitor if cardiovascular recovery between high-intensity phases is improving — this is a key indicator of aerobic adaptation. Re-check White Cell Count biomarker: if still flagged, consult GP before pushing volume further. Monitor fatigue levels from cumulative weekly training — if sleep quality drops or perceived recovery worsens, hold volume steady and do not progress load.

Week 5+ — Reassessment

Reassessment and periodisation phase. By week 5, User 2 should reassess with his physio (standard check-in given sport and training history) and review fitness scores. Consider introducing a deload week (50–60% of normal volume, same frequency) every 4–6 weeks to manage cumulative fatigue from the high-frequency programme. As User 2's cardio fitness improves, introduce Nordic Hamstring Curls as a prehab staple — starting at 3 sets of 5 reps slow eccentrics, progressing over 4–6 weeks. This is the single most evidence-based intervention for hamstring injury prevention in football players. Re-evaluate sport integration: if football frequency increases to 3x/week, reduce gym strength sessions to 2x/week accordingly to maintain guardrails. Continue monitoring White Cell Count, sleep quality, and step count — if lifestyle factors improve, overall fitness scores should increase measurably within 8–12 weeks.